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Source: *Bulletin of the History of Medicine*, NOVEMBER-DECEMBER 1972, Vol. 46, No. 6 (NOVEMBER-DECEMBER 1972), pp. 562-567

Published by: The Johns Hopkins University Press

Stable URL: <https://www.jstor.org/stable/44450832>

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DE LAGUNA'S COMMENTARIES ON HALLUCINOGENIC
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MATERIA MEDICA *

THEODORE ROTHMAN **

During the intellectual glory of the Renaissance, the irrational was fully in evidence, as it had been among the Greeks¹ and as it remains at the present time. A mixture of irrational beliefs was current among both the most sophisticated and the simplest. The persecution of witches was more widespread than it had been in the Middle Ages. A few humanists and physicians, Montaigne² and Weyer³ among others, protested but were unheeded; they were even suspected by the authorities of being wizards themselves.

The theologians supporting the irrational viewpoint were Jacobus Sprenger and Heinrich Kramer in their book *Malleus Maleficarum*.⁴ They tried to destroy witchcraft as a form of heresy. They described how witchcraft was performed and directed, and they offered remedies that would protect people from its evil powers. They concluded by demonstrating the judicial proceedings necessary in both ecclesiastical and civil courts. They showed how to convict and punish witches and wizards and all other heretics. While they correctly saw that witchcraft was another religion, they had no understanding that it represented the rites of numerous ancient religions. They only conceived witches and wizards as evil-doers, heretics, anti-Christians, who required to be destroyed.

* Translation of Renaissance Spanish into English by Walter Starkie, Litt. D., from César E. Dubler, ed., *La 'Materia Médica' de Dioscórides*, 6 vols. (Barcelona: Tipographía Emporium, 1953-1959); vol. 4: "Don Andrés de Laguna, su vida y sus obras," pp. 4-120; vol. 3: "Del Solano que engendra locura," lib. IV, cap. LXXV, fol. 420-422.

** This paper, which is based on an investigation supported in part by the Rush Research Foundation of Los Angeles, Calif., was presented at the 44th annual meeting of the American Association for the History of Medicine, Colorado Springs, Col., April 29, 1971.

¹ E. R. Dodds, *The Greeks and the Irrational* (Berkeley: Univ. of California Press, 1951), chap. 1, pp. 1-18.

² *The Essays of Michel de Montaigne*, tr. and ed. by Jacob Zeitlin from the 5th ed., 1588 (New York: Alfred A. Knopf, 1936), vol. 3, pp. 233-235.

³ Johann Weyer, *De praestigiis daemonum et incantationibus et venificiis* (Basle, 1563).

⁴ Jacobus Sprenger and Heinrich Kramer, *Malleus Maleficarum* (Nuremberg, 1494 and 1496).

They helped increase a huge growing epidemic of human destruction that included in its grasp covens of witches, thousands of innocent people, many of whom were mentally ill. Many were sadistically killed, along with followers of *il vecchio*, the ancient religions thought to have originated in the paleolithic age.⁵

A humanist who believed that witches and wizards were drug misusers and deluded was Andrés Fernández de Laguna, the celebrated Spanish physician of Emperor Charles V and Philip II.⁶ He was born and died (1499-1560) in the city of Segovia. He was also well known during the 16th century as a physician but more especially as a humanist. Although not a courtier, he made the acquaintance of Emperor Charles V while he was a young man. He dedicated his translation of Aristotle's *De Mundo* (Alcalá, 1538) to him. Philip II accepted the dedication of the translation into Spanish by de Laguna of Dioscorides' *materia medica* in 1585.

He gained his knowledge of Greek at the Sorbonne with Yacobe Tusano and Pelzo Danesio. Juan Ruellio of Paris who was the editor in 1516 of a Latin version of Dioscorides encouraged de Laguna to specialize in Greek *materia medica*. From his background one may too readily infer that de Laguna's scholarly interests were mainly humanistic. This was not entirely so, for he studied widely with a number of prominent European physicians. In 1539 when de Laguna was called in consultation on the Emperor he became a close friend of the Emperor's physician, Francisco Lopez de Villalobos. Throughout de Laguna's life he was influenced by Andrea de Mattiolo who had previously published his Latin translation of the *materia medica* of Dioscorides. Mattiolo gave special praise to de Laguna in this translation.

De Laguna was a humble man and acknowledged the help of other physicians' writings in his commentaries on Dioscorides' *materia medica*. He likewise thanked the gardeners, curers, and apothecaries who helped him understand herbals. The humanists he met in Italy were also frequently mentioned in his other works. Erasmus of Rotterdam paid tribute to him in 1548 for his humanistic studies and later Henri Estienne called him "prince of grammarians."

De Laguna's wide humanistic and medical knowledge is illustrated by his excellent Spanish translation of Dioscorides' *materia medica*.⁷ In

⁵ Margaret Alice Murray, *The Witch-Cult in Western Europe* (Oxford: Clarendon Press, 1921).

⁶ Dubler, *Dioscórides*, vol. 4, pp. 4-120.

⁷ *Ibid.*, vol. 3, lib. IV, cap. LXXV.

book IV, chapter LXXV, he describes the nightshade, classifying the different types. The nightshade he calls "a plant that engenders madness." Quoting from Dioscorides, de Laguna says, "a dram of the extract from the root when dissolved in wine produces fleeting images that please the senses, but if the dose be doubled it drives a man mad for three days; if the dose be quadrupled it can cause immediate death. With such a dangerous drug the remedy is plenty of water mixed with honey which can be vomited."

Pharmacologists today recognize the nightshade (solanaceae) as a family of herbs, shrubs, and a few trees of warm regions, chiefly tropical.³ A rank smell comes from the foliage which is typical of certain species, a smell described later by de Laguna in his commentaries. The odor is due to the presence of a number of alkaloids (including scopalamine, nicotine, and atropine), chemicals that have been used medicinally since ancient times as stimulants, narcotics, pain relievers, poisons, and antidotes for opium and snake venom. The chief drug plants of the family are the belladonna, the so-called deadly nightshade (*Atropa belladonna*), henbane (*Hyoscyamos niger*), the mandrake (*Mandragora officinum*), the Jimson weed (*Datura stamonium*), and tobacco (*Nicotiana tabacum*).

Dioscorides states according to de Laguna's translation⁹ that "if a dram of nightshade root, which produces madness, be taken in wine, it produces vain but pleasurable visions that occur as hallucinations." In his commentary de Laguna believes that, "most delusions of witches and wizards must be due to the ointments with which they anoint themselves; the excessive coldness of these ointments render them so unconscious that during their prolonged sleep a thousand stray visions so insistently appear in their brains that when they awake they confess sins they never committed." De Laguna's history and description of a case he observed help us to understand better the emotional and intellectual climate of that period. He continues,

... as a confirmation of this I can quote you an anecdote from my personal experience. When I was a salaried physician of the city of Metz, I attended the Duke Francis Lorraine, who was ill at Nancy in the year 1545. At that time a deputation happened to arrive in Metz to seek justice and retribution against two wretched old people, a husband and wife who squatted in a hermitage half a league from that city. The pair had become notorious in the whole district because it was rumored that they had burnt seed fields, killed whole herds of cattle, sucked the blood of children, and many similar infamies. On hearing such

³ *Columbia Encyclopedia*, 3rd ed., ed. by W. Bridgewater and S. Kuzer (New York: Columbia Univ. Press, 1963), p. 1505.

⁹ Dubler, *Dioscórides*, vol. 3, lib. IV, end of cap. LXXV, fol. 421-422.

accusations the Duke ordered them to be seized and tortured. They confessed that they had killed the Duke's father, Don Antonio, by giving him the grave disease which little by little consumed him. When the Duke cross-examined them, enquiring why and in what manner they had made their victim fall ill, the old man stated repeatedly that the reason he had acted thus was because on Maunday Thursday his Excellency had not selected him among the twelve poor men to have his feet washed and to receive the gift of clothing in accordance with the custom of the religious festival as he had done in other years. In consequence, the Devil noticed that he was always depressed. The Devil asked him the cause, saying to him: "If you wish to be revenged on the Duke, take this stick, and when you see him pass by your hermitage, throw the stick between the legs of the horse of the Duke and this will unseat him and end his life. On the other hand, if you do not want to kill him, but rather keep him ill, draw near him as through to beg alms and try to breathe on his face. I shall be behind you and I too shall blow over the nape of your neck, and so infect him by my breath that nobody but you could ever heal him." In this way, the wizard-hermit said he had infected the Duke, but he intended to cure him rapidly with a secret remedy which his master, the Devil, had taught him. . . . Although the Council decreed that the guilty pair should be burnt at the stake, the Duke pardoned the life of the old man, such overwhelming confidence had he in the latter's health-giving powers, and so the old woman was brutally done to death in the presence of her husband. And the old man was favored and rewarded afterwards by the Duke, although kept under close guard, until one day he was taken with his escort to the place where he had been accused, and after he had been feasted amid great jollification, next morning he was choked to death. The Duke died some days later.

De Languna asks :

What has this story to do with nightshade? Among the materials found in the hermitage of the wizard and witch was a pot full of a certain green ointment like poplar ointment [salve], whose smell was so heavy and pungent that it proved to be composed of extremely cold (ultimate grade) and soporific herbs such as hemlock, nightshade, henbane, and mandrake. Through the constable, who was a friend of mine, I obtained a good supply of the ointment and later in the city of Metz I had the wife of the public executioner anointed with it from head to foot. She through jealousy of her husband had completely lost power of sleep and had become half insane in consequence. This seemed to me to be an excellent opportunity to undertake a test of the witch's ointment. And so it turned out, for no sooner did I anoint her than she opened her eyes wide like a rabbit, and soon they looked like those of a cooked hare when she fell into such a profound sleep that I thought I should never be able to awake her. However, by means of tight ligatures and rubbing her extremities and applications of castor oil as well as euphorbia and frequent inhalations, and finally with cuppings, I was so insistent that after a lapse of thirty-six hours, I restored her to her senses and sanity. Her first words were "Why did you awaken me, badcess to you, at such an inauspicious moment? Why I was surrounded by all the delights in the world." Then turning her eyes toward her husband (he was beside her, she stinking like a corpse) and smiling at him she said: "skinflint! I want you to know that

I have put the horns on you, and with a younger and lustier lover than you." Many other strange things she said, and she wore herself out beseeching us to allow her to return to her pleasant dreams. Little by little we distracted her from her illusions, but forever after she stuck to many of her crazy notions.

From all this we may infer that all that those wretched witches do and say is caused by potions and ointments which so corrupt their memory and their imagination that they create their own woes, for they firmly believe when awake all that they had dreamed when asleep.

An additional important argument is that all those women who carried out these infamous practices confessed with one voice that they had on many occasions copulated with the Devil. And when they were asked whether they had experienced special pleasure in such connections they invariably replied that they had felt no pleasure at all because of the unbearable coldness of the Devil's private parts from which a freezing humor seemed to course through their entrails like hailstones. This sensation can only have been caused by the excessive coldness of the ointment which penetrated the marrow of their bones.

Although these scandals deserve exemplary punishment, seeing that they are associated with pacts made with the Devil, nevertheless, most of all that those witches say is mental illness, nothing else, for neither their spirit nor their body moves from the place where they collapse into unconsciousness. This is the considered opinion of the majority of theologians, and it is backed up by the decrees of a number of Holy Councils which hold that the Devil cannot operate except through natural causes. Then applying his active principles to his passive victims, and owing to his superlative knowledge and sharp wits, and because he knows the effects of such ointments, he teaches all about them to the witches, making them dream of all manner of absurd vanities. Nevertheless there are pious men who believe as gospel truth that the Devil can transform them into a hundred thousand phantoms and whisk them in body and soul through the air. With regard to all this and all the rest I cannot do better than to appeal to the just opinion of the Holy Roman Church. . . . I may add that opium is so blended in Turkey that when swallowed it also causes very pleasant dreams accompanied by all the happiness one could possibly desire and it appears that this agrees with what Dioscorides says concerning the nightshade.

De Laguna's translation into Spanish of the Greek of Dioscorides was one of his most important humanistic contributions. By having translated this work into Spanish he helped advance Spanish medicine further and it was able to develop its own *materia medica*. His personal commentaries included in the translation demonstrate an astute power of original clinical observation. He was perhaps the first physician to correlate the drug use of the nightshade with the rituals of witchcraft. This drug had a similar hallucinogenic effect to LSD, and de Laguna noted that these were pleasurable experiences and were hallucinatory and delusional, produced by the ointments they prepared for the occasional festivals of the witches and wizards. He proved that these hallucinogenic drugs, when used on other subjects, would have a similar effect. He emphasized that the drugs

produced increased suggestability and that the confessions wrung from the drug users represented the delusional speech of a deranged mind.

While he paid full respect to all the beliefs of Roman Catholic theologians and supported some of the contemporary irrationalisms, his new medical and psychiatric perspectives brought new light and direction on witchcraft, and a scientific attitude towards these deviants. This rational approach triumphed in the nineteenth century. Meanwhile, there are a multitude of other irrationalities being practiced at the present time, replacing the cruelties towards supposed witchcraft by bringing highly sophisticated death to multitudes of other people for similar disguised sado-masochistic abstractions.

ACKNOWLEDGMENT

I should like to express my thanks to the Rector of the University of Barcelona School of Medicine, Dr. F. G. Valdecasas, for making available to me chapter 75, pages 471-472, of de Laguna's original book and for his helpful suggestions.